



Kire Kolaroski

Senior Machine Learning Engineer

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SUMMARY

ML engineer with 8 years building production ML systems end to end — from ambiguous problem brief to deployed platform. Owned or co-owned three full-lifecycle platforms across time-series forecasting, computational biology, and medical imaging — the last five years fully remote with US clients. Specialises in problem spaces where the modelling approach isn't defined in advance: translating domain problems into systems that biologists, clinicians, and analysts actually use, with expert feedback loops built in.

EXPERIENCE

Machine Learning Engineer / Data Scientist

FatRat · 2021 – Present · Skopje *Fully remote — async delivery to US-based clients across a 6–9 hour time zone gap*

Lead engineer on ML product engagements for US clients. Owned projects end to end, from problem definition and data sourcing through architecture, modelling, and delivery.

Biological network analysis and drug property platform — US pharmaceutical client — Lead

- Architected and built a platform wrapping the Reactome pathway database, ingesting its analysis output to identify key intervention nodes required to return a perturbed biological network to equilibrium.
- Designed and implemented the heuristic search algorithms underlying node identification; results were reviewed and validated by client-side biologists in an iterative feedback loop.
- Grew the engagement from an initial polygenic risk scoring brief into a full analysis platform as the client's requirements developed.
- Built a parallel pipeline predicting molecular properties from SMILES representations, covering ADMET-style endpoints.

Mitosis detection and expert-in-the-loop labelling platform — biotech client — Contributor

- Worked on complete detection pipeline: whole-slide scan ingestion, 512×512 tiling, and a semantic segmentation model for detecting mitotic figures.
- Identified label quality as the binding constraint on accuracy and built a labelling platform allowing biologists and clinicians to correct model predictions directly.
- Helped establish an active-learning loop feeding corrected labels back into training. The dataset reached **200,000 labelled samples**, with model accuracy improving on every iteration.

Machine Learning Engineer

causaLens · 2018 – 2020 · London, UK *On-site London, transitioned to remote in 2020*

First employee at a time-series forecasting startup. Scope spanned research, core platform engineering, client modelling, and hiring.

Automated time-series forecasting platform — Contributor

- Core contributor to a general-purpose forecasting platform covering data loading, cleaning, feature generation, baseline model generation, hyperparameter optimisation, and automated model summary reporting.
- Implemented Cython components for high-throughput streaming data generation feeding the forecasting engine.
- Designed and built the model scoring and model explainability subsystems.
- Implemented HDF5 and CSV ingestion layers.

Client modelling and delivery

- Worked directly with financial institutions and other clients to translate business problems into forecasting models; delivered solutions across equities, shipping, and gambling domains.
- Designed and implemented neural network architectures for time-series prediction.
- Ran technical interviews for data scientist and machine learning engineer hires.

Research Intern

Macedonian Academy of Sciences and Arts, Computer Science Department · 2017

- Researched information retention in the aggregation of multiplex networks, using entropy-based measures to quantify information loss after aggregation.

SKILLS

Languages — Python, Cython, SQL, JavaScript

Machine learning — PyTorch, scikit-learn, time-series forecasting, semantic segmentation, hyperparameter optimisation, AutoML, model explainability, heuristic and graph algorithms

Data and infrastructure — Kubernetes, Docker, PostgreSQL, AWS

Backend and web — Django, FastAPI, React

Domains — Financial time series, computational biology, cheminformatics, medical imaging

EDUCATION

BSc, Computer Science and Engineering · Ss. Cyril and Methodius University, Skopje · 2013 – 2017 Mathematics-focused track: calculus, linear algebra, probability, statistics, natural language processing, pattern recognition. (9.1/10)